



CSE (DATA SCIENCE)

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Title of the project:

**JEPA-LLM Tuning Framework for Combating GST and Sales Tax Input
Tax Credit Fraud Pattern Detection**

Project Batch No:A5

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Abstract: Goods and Services Tax (GST) and Sales Tax Input Tax Credit (ITC) fraud have become a major concern in modern tax administration due to the increasing use of fake invoices, circular trading, shell companies, and fraudulent credit claims. These frauds cause substantial revenue leakage and weaken the integrity of the taxation system by enabling taxpayers to claim illegitimate credits without genuine business transactions. Traditional fraud detection methods, which rely heavily on manual auditing, rule-based checks, and post-fraud investigation, are often slow, reactive, and insufficient for uncovering large-scale hidden fraud networks. In real-world GST ecosystems, fraud patterns are not always directly visible in individual transactions; instead, they emerge through complex behavioural, temporal, and relational patterns spread across multiple invoices, taxpayers, and filing cycles. This creates a strong need for an intelligent, scalable, and explainable fraud detection framework capable of learning hidden patterns from large volumes of tax data.

To address this challenge, this project proposes a JEPA-LLM Tuning Framework for Combating GST and Sales Tax Input Tax Credit Fraud Pattern Detection. The framework combines domain-specific GST fraud knowledge, self-supervised representation learning, fraud scoring, and natural-language explanation to identify suspicious ITC claims more effectively. The proposed system is inspired by three major directions: GST fraud and forensic accounting analysis, hybrid AI-based financial fraud detection, and explainable multi-layer fraud assessment. At the domain level, the framework targets major GST/ITC fraud behaviours such as fake invoice generation, abnormal input credit claims, shell-entity participation, circular trading loops, invoice without actual movement of goods or services, and coordinated fraud among linked taxpayers.

A core contribution of this project is the integration of Joint Embedding Predictive Architecture (JEPA) as the primary pattern-learning component. In this framework, JEPA is used as a self-supervised representation learning module that learns hidden behavioural structures from largely unlabeled GST data. Instead of depending only on fully labelled fraud examples, JEPA models the normal relationships among invoices, taxpayers, filing history, and transaction flows, and then learns compact predictive embeddings that

represent expected GST behaviour. When a taxpayer, invoice sequence, or transaction network deviates significantly from these learned behavioural patterns, the system flags it as potentially suspicious. This makes JEPA particularly valuable for GST fraud detection because real tax datasets often contain very limited confirmed fraud labels, while fraudsters continuously change their tactics to avoid rule-based detection. By learning behavioural regularities and deviations, JEPA helps capture subtle fraud signatures such as unusual spikes in ITC claims, repeated invoice cycles, suspicious buyer–seller relationships, circular credit movement, and shell-company-linked transaction behaviour.

The proposed framework follows a multi-layer fraud detection strategy by combining three categories of features: invoice-level features, taxpayer behaviour features, and transaction/network-level features. Invoice-level features may include invoice amount, tax value, credit claimed, filing frequency, duplicate invoice indicators, and date inconsistencies. Taxpayer behaviour features capture patterns such as sudden increases in ITC utilization, mismatch between outward and inward tax declarations, refund anomalies, and irregular filing habits. Network-level features model buyer–seller relationships, repeated entity interactions, shared contact information, shell-company clusters, and circular trading loops. JEPA encodes these heterogeneous patterns into a unified latent representation, which is then passed to a fraud scoring module for classification or anomaly detection.

To enhance interpretability and usability, the framework also integrates LLM tuning as an explanation and decision-support layer. Once suspicious entities or transactions are identified, the LLM generates human-readable fraud summaries, highlighting the probable reasons behind each alert. For example, it can explain whether a taxpayer was flagged due to abnormal ITC growth, repeated transactions with linked firms, suspicious invoice chains, or inconsistent filing behaviour. This improves transparency and makes the system more useful for auditors, investigators, and tax officers who require understandable evidence rather than raw anomaly scores alone.

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